

Question number	Answer	Additional guidance	Mark
)(i)	An answer that includes the following points: • correct location of the Bowman's capsule (1)		(1)

Question number	Answer	Additional guidance	Mark
)(ii)	An answer that includes the following point: D is the correct answer • A is not the correct answer as collecting duct collects urine from the nephrons • B is not the correct answer as the distal tubule is responsible for salt and water reabsorption • C is not the correct answer as the loop of Henle is responsible for salt and water reabsorption		(1)

Question number	Answer	Additional guidance	Mark
)(iii)	An answer that includes the following point: - B is the correct answer - A is not the correct pathway - C is not the correct pathway - D is not the correct pathway		(1)

Question number	Answer	Additional guidance	Mark
)	An explanation that includes four of the following points: - flow of filtrate in tubules in opposite direction (1) - because sodium ions actively transported out of the ascending limb (1) - therefore water out from descending limb (and solutes from ascending limb) (1) - due to increase in concentration gradient between tubular fluid and interstitial space (1)		(4)

Question number	Answer	Additional guidance	Mark
(c)	A description that includes four of the following points: - water potential of blood monitored by osmoreceptors in hypothalamus (1) - release of ADH from (hypothalamus) pituitary gland (1) (in blood to kidneys)where ADH acts on {collecting ducts / distal convoluted tubul} (1) - increase in permeability to water of cell membranes (1) - more water reabsorbed so water potential of blood back to normal range (1)	Accept insertion of aquaporins into cell membranes (of CT and DCT)	(4)